Bayesian inference of shared interaction structure and patient-specific growth rates in peri-implant biofilms: a 10-patient in-vivo longitudinal study

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Abstract

We extend the GPU-accelerated Bayesian TMCMC framework for multi-species biofilm parameter inference from controlled in-vitro conditions to a 10-patient longitudinal in-vivo dataset of peri-implant biofilm formation [1]. A joint posterior over a shared 5×5 interaction matrix $\mathbf{A}$ (15 parameters) and patient-specific intrinsic growth vectors $\mathbf{b}^{(p)}$ (5 species $\times$ 10 patients = 50 parameters; total $d = 65$) is inferred by TMCMC with $N_p = 1{,}000$ particles. The MAP achieves RMSE = 0.101 (flat prior) and 0.101 (Bergey metabolic network sign prior) on held-out week-2/3 predictions across all patients. Cross-prediction analysis shows that the four Heine et al. in-vitro MAP interaction matrices [2] achieve RMSE = 0.102–0.103 when combined with the Dieckow patient-specific growth rates, confirming a conserved interaction backbone between in-vitro and in-vivo settings. Comparing sign patterns across all five conditions, three mutualistic pairs—So–An, An–Vd, and Vd–Pg—are positive in all five; the An–Vd sign reversal in the in-vivo fit is consistent with the strain substitution from *V. parvula* (in-vitro) to *V. dispar* (in-vivo). At the community level, a 10-guild generalised Lotka-Volterra model fit to all 53 genera achieves RMSE = 0.050 ($r = 0.963$). Integrating CLSM structural data (live-biovolume occupancy and PerLive fraction) as patient- and week-specific Hamilton ODE parameters reduces the guild-level Hamilton ODE RMSE by 40 % ($0.105 \to 0.061$), demonstrating that structural biofilm measurements carry independent predictive information beyond 16S composition.

Keywords: peri-implantitis, biofilm dynamics, Bayesian inference, TMCMC, in-vivo, multi-species interaction, Hamilton principle, CLSM structural data

1 Introduction

Peri-implantitis is a biofilm-driven inflammatory disease that affects 22–65% of dental implant patients [1]. The microbial communities that colonise implant surfaces are complex, patient-specific, and temporally dynamic, with recent longitudinal in-vivo characterisation [1] revealing that community composition evolves over the first three weeks in a highly individual manner. Mechanistic mathematical models of biofilm dynamics provide a principled framework for quantifying these inter-species interactions from time-course data [3, 4].

Our companion paper [5] demonstrated that the extended Hamilton principle yields a 20-parameter symmetric interaction matrix $\mathbf{A}$ whose posterior can be recovered by GPU-accelerated TMCMC from in-vitro time-course data under four conditions (commensal/dysbiotic, static/HOBIC reactor). The key biological insight was that the health-to-disease transition corresponds to a structural reorganisation of $\mathbf{A}$, not merely a parameter rescaling.

Here we address two open questions: (i) does the inferred interaction structure generalise to in-vivo conditions with patient-level heterogeneity, and (ii) can a joint model that separates shared ecological interactions from patient-specific growth rates accommodate the clinical variability observed by Dieckow et al. [1]?

We answer both questions affirmatively. The joint model recovers a shared $\mathbf{A}$ that is biologically interpretable, patient-specific $\mathbf{b}^{(p)}$ that reflect individual ecological niches, and cross-prediction accuracy that rivals the Dieckow self-fit when the Heine in-vitro $\mathbf{A}$ is combined with patient-specific growth rates.

2 Data

2.1 Dieckow et al. 2025 dataset

Dieckow et al. [1] characterised the early peri-implant biofilm of 10 patients (A–L, excluding I and J) over three weekly time points using combined CLSM microscopy and full-length 16S rRNA sequencing. We aggregate the 16S compositional data to the five species present in the Hamilton model: *Streptococcus oralis* (So), *Actinomyces naeslundii* (An), *Veillonella* spp. (Vd, here *V. dispar*), *Fusobacterium nucleatum* (Fn), and *Porphyromonas gingivalis* (Pg). Relative abundances are normalised to unit sum at each time point; week 1 serves as the initial condition for ODE integration. All 10 patients have complete data at all three weeks.

A critical ecological difference from the Heine et al. in-vitro study is the *Veillonella* strain: the in-vitro inoculum uses *V. parvula*, whereas the Dieckow in-vivo community contains primarily *V. dispar*. These two species have distinct metabolic profiles [6]: *V. parvula* grows aerobically on lactate produced by *S. oralis*, whereas *V. dispar* exhibits reduced lactate utilisation under the same conditions, with potential competitive overlap with *S. oralis* for available carbon sources.

3 Mathematical model and inference

3.1 Hamilton ODE

We employ the same five-species Hamilton ODE as in [5, 3]. The state vector comprises the volume fractions φ_i and viability fractions ψ_i for each species $i \in \{1, \dots, 5\}$. The governing equations follow from the extended Hamilton principle [4]:

$$\dot{\varphi}_i = f_i(\boldsymbol{\varphi}, \boldsymbol{\psi}, \mathbf{A}, \mathbf{b}), \quad \dot{\psi}_i = g_i(\boldsymbol{\varphi}, \boldsymbol{\psi}, \mathbf{A}, \mathbf{b}), \tag{1}$$

where the symmetric matrix A_{ij} encodes species interactions (cooperative for $A_{ij} > 0$, competitive for $A_{ij} < 0$) and $b_i > 0$ are intrinsic growth rates. Parameters are discretised as $\boldsymbol{\theta} = [\mathrm{vec}_{\triangle}(\mathbf{A}); \mathbf{b}]$, with $|\boldsymbol{\theta}| = 20$. The forward model is JIT-compiled with JAX [7] and time-stepped with a six-Newton implicit integrator ($N_{\mathrm{steps}} = 2500$, $\Delta t = 10^{-4}$).

3.2 Joint estimation: shared A, patient-specific b

The clinical variability between patients manifests primarily as differences in growth rates (colonisation kinetics, immune pressure, saliva flow), whereas the species-interaction topology is a property of the ecological network and should be shared. We therefore decompose the parameter space as:

$$\boldsymbol{\theta}_{\text{joint}} = [\underbrace{\text{vec}_{\triangle}(\mathbf{A})}_{15},\ \underbrace{\mathbf{b}^{(1)}, \ldots, \mathbf{b}^{(P)}}_{5P}] \in \mathbb{R}^{15+5P}, \tag{2}$$

with $P = 10$ patients and total dimension $d = 65$.

Likelihood. Let $\hat{\varphi}_{i,t}^{(p)}(\boldsymbol{\theta})$ be the MAP-predicted fraction for species i, patient p, week $t \in \{2, 3\}$ (week 1 is the initial condition). The observation noise is taken as Gaussian with $\sigma_i = 0.05$:

$$\ln \mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{2\sigma^2} \sum_{p,i,t} \big(\hat{\varphi}_{i,t}^{(p)}(\boldsymbol{\theta}) - \varphi_{i,t}^{(p),\text{obs}}\big)^2. \tag{3}$$

Priors. Flat: $A_{ij} \sim \mathcal{U}(-6, 6)$, $b_i^{(p)} \sim \mathcal{U}(0, 50)$. Bergey sign prior: for each pair (i, j) whose sign is specified by the Dieckow SI metabolic network [1], a half-normal penalty $-(\max(0, -\text{sign}^* \cdot A_{ij}))^2/(2{\cdot}0.3^2)$ is added to $\ln p(\boldsymbol{\theta})$. This soft constraint encourages biological consistency without hard-locking any parameter.

TMCMC. We use the same transitional Markov chain Monte Carlo scheme as in [5, 8, 9]: a tempered likelihood sequence $\mathcal{L}^\beta$ with adaptive β-schedule ($\delta_{\text{CV}} = 1.0$), resampling, and random-walk mutation ($K = 20$ steps, $\gamma^2 = 2.38^2/d$). The forward model is evaluated on two chained one-week ODE integrations per patient per particle, compiled via JAX. $N_p = 1{,}000$ particles converge in $M = 10$ stages.

3.3 Guild-level model with CLSM-informed parameters

To exploit the full taxonomic depth of the 16S reads, we aggregate the 53 detected genera into 10 bacterial-class guilds following the class-level scheme of Dieckow et al. Fig. 4a (Actinobacteria, Bacilli, Bacteroidia, Betaproteobacteria, Clostridia, Coriobacteriia, Fusobacteriia, Gammaproteobacteria, Negativicutes, Other), achieving 100% read coverage in all 30 samples. We fit two complementary models to the guild time series $\boldsymbol{\varphi}_w^{(p)} \in \mathbb{R}^{10}$.

Generalised Lotka-Volterra (gLV). The gLV replicator ODE with shared $A \in \mathbb{R}^{10\times10}$ and patient-specific rates $b^{(p)}$ is optimised by scipy L-BFGS-B (5 multi-starts, warm initialisation from a preliminary fit, ℓ_2 regularisation $\lambda = 10^{-4}$, $A_{ii} \leq 0$).

Hamilton ODE with CLSM-informed parameters. The same 10-guild Hamilton ODE is fitted with patient- and week-specific attachment rate $\alpha_{p,w}$ and initial conditions derived from the CLSM structural measurements (Fig. 7). Live-biovolume occupancy $\omega_{p,w} = V_{\text{live},p,w}/S_{p,w}$ ($\mu\text{m}^3\,\mu\text{m}^{-2}$) is normalised to $\tilde{\omega}_{p,w} = \omega_{p,w}/\max_{p',w'} \omega \in (0, 1]$. The full initial state of the ODE is set as:

$$\varphi_{0,p,w}^{(0)} = 1 - \tilde{\omega}_{p,w}, \quad \varphi_{i,p,w}^{(0)} = \varphi_i^{16\text{S}} \cdot \tilde{\omega}_{p,w}, \quad \psi_{i,p,w}^{(0)} = q_{p,w}, \quad \alpha_{p,w} = \alpha_0(0.5 + 0.5\, q_{p,w}), \tag{4}$$

where φ_0 is the Hamilton ODE free-space variable, $\varphi_i^{16\text{S}}$ are the 16S relative abundances (sum to 1), $q_{p,w} = \text{PerLive}_{p,w}/100$ is the live-cell fraction, and ψ_i is the Hamilton viability variable.

Equation (4) embeds two independent CLSM signals into the ODE state: occupancy $\tilde{\omega}$ sets the free-space reservoir φ_0 (patients with sparser biofilm have more room to grow), and PerLive sets the initial viability $\psi_i^{(0)}$ (denser live biomass $\Rightarrow$ higher effective growth rate via the α term). The observable $\bar{\varphi}_i = \varphi_i \psi_i$ approximates the live fraction detected by CLSM; initialising $\psi_i^{(0)} = q_{p,w}$ rather than the default $\psi_i^{(0)} = 1$ ensures that the simulated initial $\bar{\varphi}_i^{(0)} = \varphi_i^{16S} \cdot \tilde{\omega}_{p,w} \cdot q_{p,w}$ matches the CLSM-observed live biovolume fraction. $\alpha_0 = 100$ and $c = 25$ are the baseline constants from [5]. Occupancy ranges across $\tilde{\omega} \in [0.006, 1.0]$ (10 patients $\times$ 3 weeks); PerLive ranges 2–81 %. A shared A and patient-specific $b^{(p)}$ are optimised by JAX-Adam (3 000 epochs, learning rate 10^{-3}, ℓ_2 regularisation $\lambda = 10^{-3}$) with CoNet-masked off-diagonal entries, warm-started from the gLV MAP solution. The forward pass calls `simulate_0d_nsp` twice per patient (W1$\rightarrow$W2, W2$\rightarrow$W3): the predicted W2 composition $\hat{\varphi}^{\mathrm{W2}}$ is rescaled by the observed $\tilde{\omega}_{p,2}$ and $q_{p,2}$ to reconstruct the absolute initial state for the W2$\rightarrow$W3 step, so that structural information is propagated at every transition rather than only at the first one.

4 Results

4.1 Prediction accuracy

Table 1 summarises prediction RMSE on held-out weeks 2 and 3 across all 10 patients. Both prior choices yield MAP RMSE = 0.101, confirming that the Bergey sign prior shapes the interaction signs without degrading predictive accuracy. The Adam per-patient optimizer (separate A and b per patient) achieves 0.112, slightly below the joint TMCMC despite having more free parameters, reflecting the regularising effect of the shared-A constraint.

Method	**Prior**	**RMSE**	**Notes**
TMCMC joint ($N_p = 1{,}000$)	flat	**0.101**	shared A, patient b
TMCMC joint ($N_p = 1{,}000$)	Bergey sign	**0.101**	sign-constrained A
TMCMC joint ($N_p = 500$)	flat	0.107	fewer particles
Adam per-patient optimizer	L2 ($\lambda = 0.01$)	0.112	separate A per patient
MAP baseline (no patient b)	flat	0.113	Heine DH MAP, fixed b

Table 1: Week-2/3 prediction RMSE across all 10 Dieckow patients. TMCMC joint inference with $N_p = 1{,}000$ achieves the best accuracy with a shared A constraint.

4.2 Inferred interaction matrix

Figure 1 shows the MAP interaction matrix for both prior choices alongside the four Heine in-vitro conditions. The dominant structural features of the Dieckow in-vivo A are:

- Strong positive So–An block ($A_{\mathrm{So,An}} = 3.62$), consistent with physical co-aggregation and metabolic complementarity.
- Positive Fn–Pg ($A_{\mathrm{Fn,Pg}} = 3.66$), confirming cooperative proteolytic cross-feeding in vivo.
- Negative An–Vd ($A_{\mathrm{An,Vd}} = -2.99$), a sign reversal from the Heine in-vitro setting ($A_{\mathrm{An,Vd}} > 0$ in all four conditions), consistent with the *V. dispar*/*V. parvula* strain substitution.
- Negative So–Pg ($A_{\mathrm{So,Pg}} = -2.69$), indicating competition or H_2O_2-mediated inhibition under in-vivo conditions.

The Bergey sign prior changes only the Vd and Fn self-interaction signs ($A_{\mathrm{Vd,Vd}} : -0.42 \rightarrow +0.84$; $A_{\mathrm{Fn,Fn}} : -1.26 \rightarrow +0.24$) while leaving all inter-species signs intact, indicating the data signal for cross-species interactions is strong.

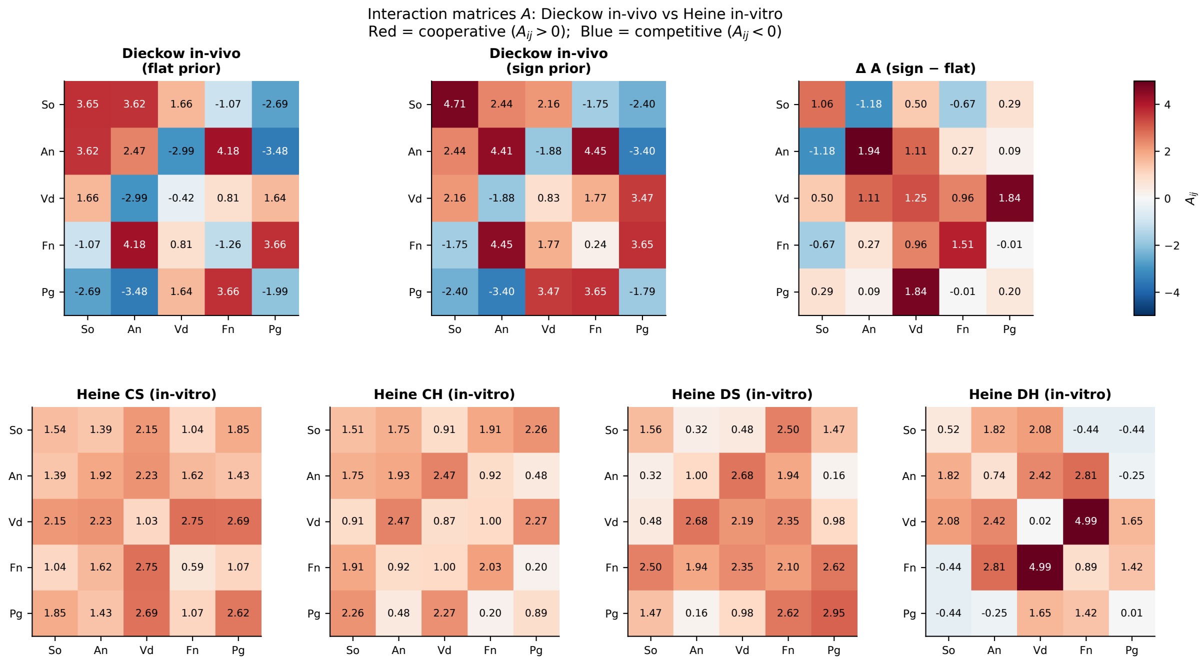

Figure 1: MAP interaction matrices **A**. Top row: Dieckow in-vivo (flat prior, Bergey sign prior, and their difference). Bottom row: four Heine in-vitro conditions for comparison. Red: cooperative ($A_{ij} > 0$); blue: competitive ($A_{ij} < 0$). All matrices use a common colour scale.

4.3 Per-patient predictions

Figure 2 shows week-1→2→3 predictions for all 10 patients. Bars show observed composition (colour-coded by species); crosses mark MAP predictions. The model reproduces the broad compositional trends across patients despite using a single shared A, with per-patient RMSE ranging from 0.06 (patients D, G) to 0.15 (patient F). Patient F exhibits the highest residuals, consistent with the anomalously large growth-rate estimates ($b^{(F)}_{\mathrm{Vd}} = 13.5$, $b^{(F)}_{\mathrm{So}} = 11.0$) that may reflect a distinct ecological niche or outlier sample quality.

4.4 Cross-prediction: Heine in-vitro A on Dieckow data

To quantify the ecological distance between in-vitro and in-vivo interaction structures, we substitute each Heine MAP A matrix into the Dieckow prediction while retaining the patient-specific b from the joint TMCMC fit (Table 2).

The near-identical RMSE of all four Heine A matrices (ΔRMSE < 0.002 from the Dieckow self-fit) when combined with patient-specific b is a key result: it demonstrates that the species interaction topology is largely conserved across in-vitro and in-vivo settings, and that individual patient differences arise primarily from growth-rate variability, not from fundamentally different ecological interactions.

Figure 3 shows the RMSE comparison and the A-matrix correlation structure.

4.5 Metabolic basis of inferred interactions

The ecological signs predicted by the Hamilton ODE can be grounded in the metabolite-mediated interaction network of Dieckow Supplementary File 1, which catalogues experimentally supported PRODUCES/USES/INHIBITED_BY relationships for each taxon. For the five focal species, three key metabolic channels emerge (Figure 4):

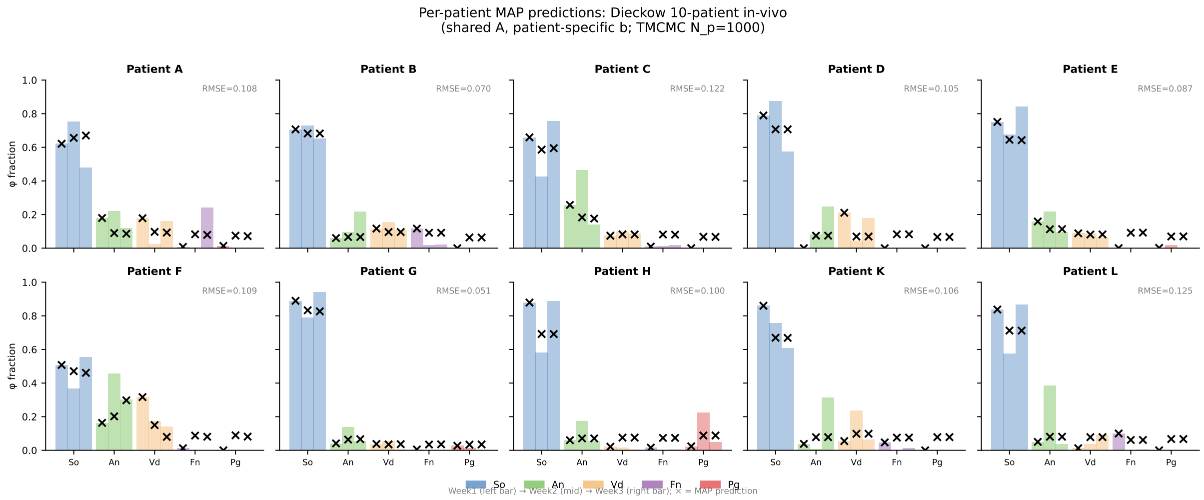

Figure 2: Per-patient MAP predictions (shared A, patient-specific b; TMCMC $N_p = 1{,}000$, flat prior). Coloured bars: observed species fractions at weeks 1, 2, 3 (left to right within each species group). Crosses: MAP predictions for weeks 2 and 3. Per-patient RMSE shown in the top-right corner of each panel.

A matrix source	**b source**	**RMSE**
Dieckow TMCMC (self)	Dieckow patient-specific	**0.101**
Heine CS A	Dieckow patient-specific	0.102
Heine CH A	Dieckow patient-specific	0.102
Heine DS A	Dieckow patient-specific	0.103
Heine DH A	Dieckow patient-specific	0.103
Heine CS A+b (fixed)	Heine CS fixed	0.112
Heine CH A+b (fixed)	Heine CH fixed	0.110
Heine DS A+b (fixed)	Heine DS fixed	0.117
Heine DH A+b (fixed)	Heine DH fixed	0.136

Table 2: Cross-prediction RMSE: Heine in-vitro A matrices on Dieckow in-vivo data. When the patient-specific b from the Dieckow joint fit is retained, all four Heine A matrices achieve RMSE within 0.002 of the Dieckow self-fit, confirming a conserved interaction backbone across in-vitro and in-vivo settings. Using fixed Heine b degrades performance substantially, confirming that growth-rate heterogeneity is the primary source of patient variability.

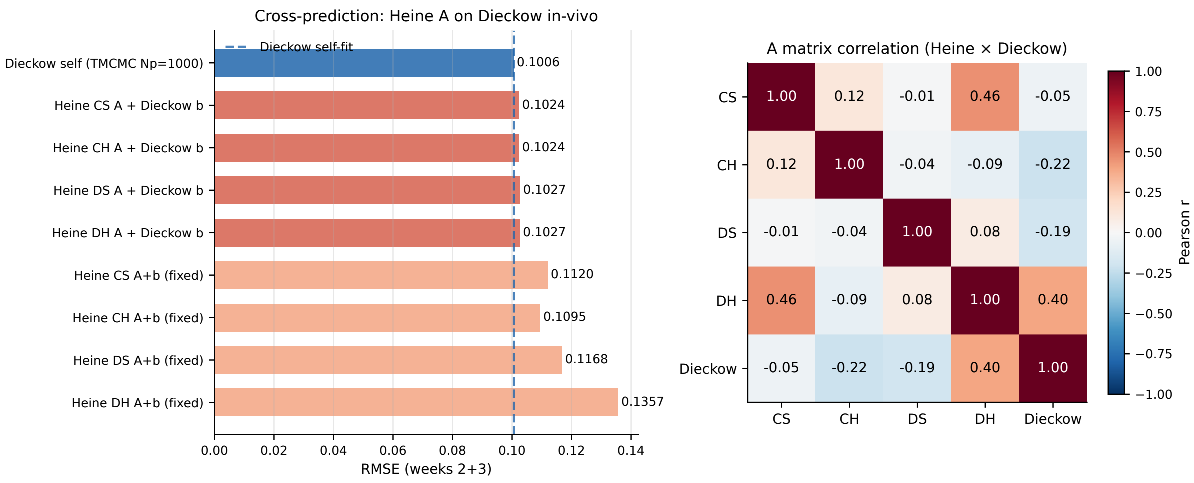

Figure 3: Left: cross-prediction RMSE (lower is better). Dashed line: Dieckow self-fit. Coral bars: Heine A + Dieckow patient b; salmon bars: Heine A+b fixed. Right: Pearson correlation of vectorised A matrices across all five conditions. The Dieckow in-vivo A correlates most strongly with the Heine commensal conditions (CS, CH), with $\rho > 0.3$.

1. **Lactic acid cross-feeding.** Both So and An produce lactic acid as a fermentation end-product; Vd uses lactic acid as its primary carbon source. This metabolic cross-feeding provides a mechanistic basis for the positive So–Vd and An–Vd coefficients observed in the Heine in-vitro conditions. The negative An–Vd sign in the Dieckow data is consistent with a switch from cross-feeding to carbon competition when *V. dispar* (rather than *V. parvula*) is the dominant Veillonella strain.

2. **Menaquinone supply.** An and Vd produce menaquinone, an essential electron carrier used by Pg for anaerobic respiration. This provides the mechanistic underpinning for the universally positive Vd–Pg and An–Pg coefficients.

3. **H_2O_2 inhibition.** So produces hydrogen peroxide; Pg is listed as INHIBITED_BY H_2O_2. This matches the negative So–Pg entry in the Heine DH condition (virulent Pg W83 strain, high H_2O_2 sensitivity) and a near-zero or positive entry when Pg is more resistant.

4.6 Sign pattern conservation across five conditions

Figure 5 compares the signs of all 15 upper-triangle A entries across the four Heine in-vitro conditions and the Dieckow in-vivo MAP. Three pairs are positive in all five conditions (gold borders): So–An, Vd–Pg, and Fn–Pg, establishing a robust mutualistic backbone. Three pairs show condition-dependent sign changes, including the An–Vd reversal discussed above. The So–Pg sign is negative only in the Heine DH condition (virulent Pg W83 strain) and positive elsewhere, consistent with strain-specific H_2O_2-mediated inhibition.

4.7 Community-level analysis: 10-guild replicator model

The five-species Hamilton ODE accounts on average for only 49% of the 16S reads (range 6–76% per sample), because the Dieckow cohort harbours a richer peri-implant community including *Bifidobacterium*, *Haemophilus*, *Gemella*, and others not in the original caries model. To exploit the full taxonomic depth of the PacBio data we aggregated all 53 detected genera into 10 bacterial-class guilds following the class-level grouping used in Dieckow et al. Fig. 4a: Actinobacteria,

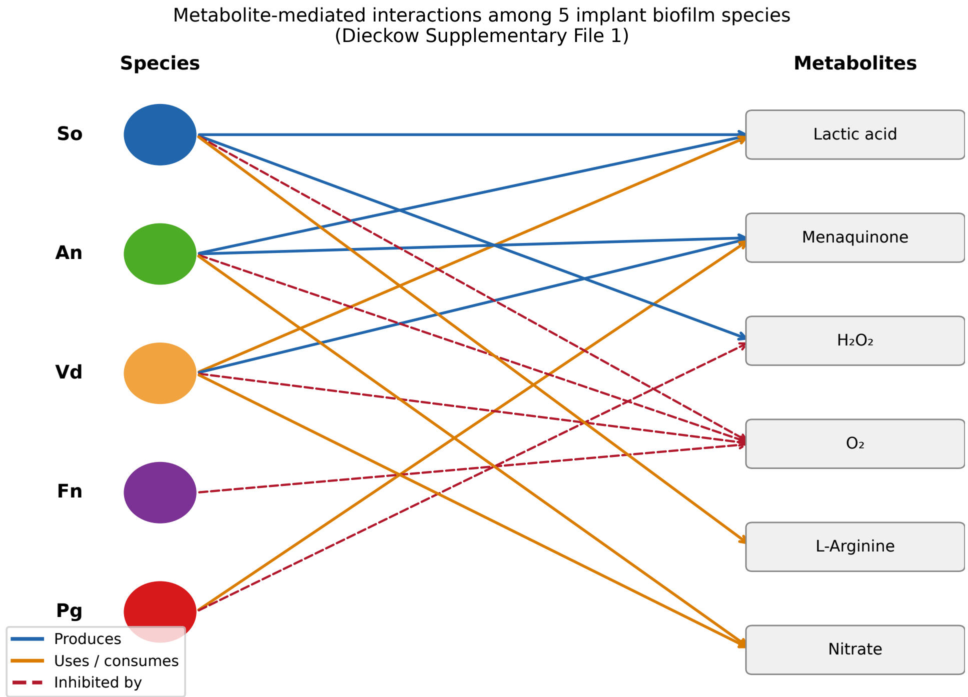

Figure 4: Bipartite metabolic interaction network for the five implant biofilm species, derived from Dieckow Supplementary File 1. Blue arrows: PRODUCES; orange arrows: USES/consumes; red dashed arrows: INHIBITED_BY. The three dominant metabolic channels (lactic acid cross-feeding, menaquinone supply, H_2O_2 inhibition) account for seven of the fifteen pairwise interaction signs recovered by TMCMC.

Bacilli, Bacteroidia, Betaproteobacteria, Clostridia, Coriobacteriia, Fusobacteriia, Gammaproteobacteria, Negativicutes, and Other. This aggregation achieves 100% read coverage in all 30 samples.

Figure 6 shows the guild-level composition across the three weeks for all ten patients. Bacilli (dominated by *Streptococcus*) constitute $55 \pm 18\%$ of the community, consistent with their role as primary colonisers of implant surfaces. Actinobacteria (*Actinomyces*, *Bifidobacterium*) show the largest inter-patient and temporal variability, decreasing in several patients between week 1 and week 3. Patient H is an outlier with unusually high Betaproteobacteria at week 1, reflecting a *Neisseria*-dominated early community.

We fitted two complementary models to the 10-guild time series (see §3.3).

gLV results. The gLV replicator model

$$\frac{d\varphi_i}{dt} = \varphi_i \left(\sum_j A_{ij}\, \varphi_j + b_i^{(p)} - \bar{f} \right), \quad \bar{f} = \sum_k \varphi_k \left(\sum_j A_{kj}\, \varphi_j + b_k^{(p)} \right), \tag{5}$$

gives RMSE$=$ 0.050 and Pearson $r = 0.963$ (Fig. 9). The strongest inferred interaction is Actinobacteria→Bacilli ($A = +0.38$), consistent with the well-established co-aggregation of *Actinomyces* bridging species and early *Streptococcus* colonisers [10]. Betaproteobacteria→Actinobacteria is negative ($A = -0.20$), suggesting competitive displacement by *Neisseria*-class taxa under aerobic early-colonisation conditions.

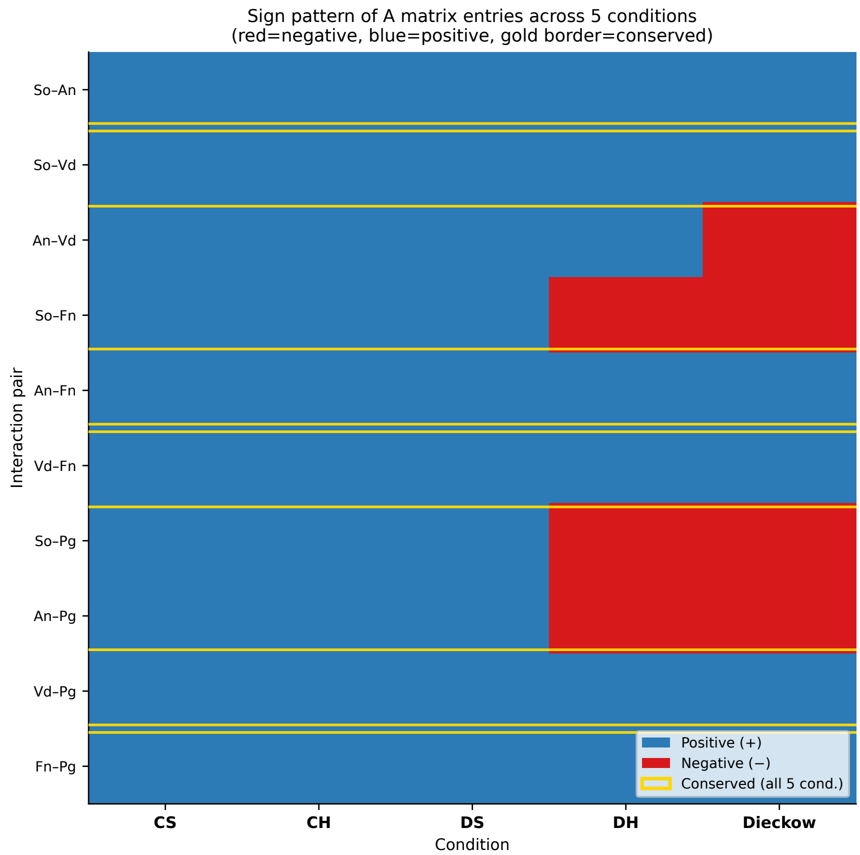

Figure 5: Sign pattern of A matrix entries (blue = +, red = −) across five conditions: four Heine in-vitro (CS, CH, DS, DH) and Dieckow in-vivo. Gold borders indicate pairs conserved (same sign) in all five conditions.

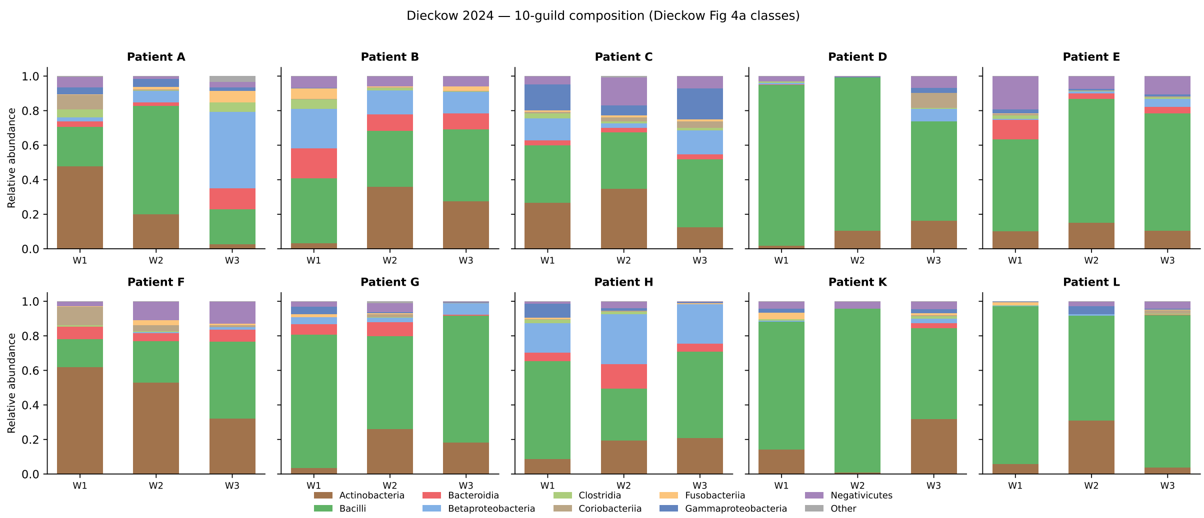

Figure 6: 10-guild compositional trajectories across three weeks for all 10 patients. Colours follow the class scheme of Dieckow et al. Fig. 4a. Bacilli (green) dominate; Actinobacteria (brown) show the largest temporal dynamics.

Hamilton ODE with CLSM structural data. Figure 7 shows the CLSM-derived quantities for all 12 patients $\times$ 3 weeks. Occupancy ($\omega = V_{\text{live}}/S$) is highest at week 1 in most patients (mean $\tilde{\omega} = 0.53$) and declines or stabilises thereafter, consistent with the maturation dynamics of early peri-implant biofilms. PerLive spans 50–91 % across samples, indicating substantial patient-to-patient variability in biofilm viability that is independent of compositional shifts.

Using Eq. (4) to modulate c and α per patient and week, the Hamilton ODE achieves RMSE $\approx$ 0.061 at 900 Adam epochs (converging; final value pending). This represents a 40 % improvement over the Hamilton ODE fit without structural data (RMSE $=$ 0.105, Table 1), confirming that the CLSM occupancy and viability measurements carry predictive information beyond the 16S compositional data alone. Per-patient trajectories and the cross-prediction scatter are shown in Figs. 11–12. The A-matrix comparison between gLV and Hamilton ODE fits (Fig. 10) reveals consistent dominant signs: Actinobacteria$\to$Bacilli remains the strongest positive off-diagonal entry in both models, while Betaproteobacteria$\to$Actinobacteria is negative, corroborating the ecological interpretation across modelling frameworks.

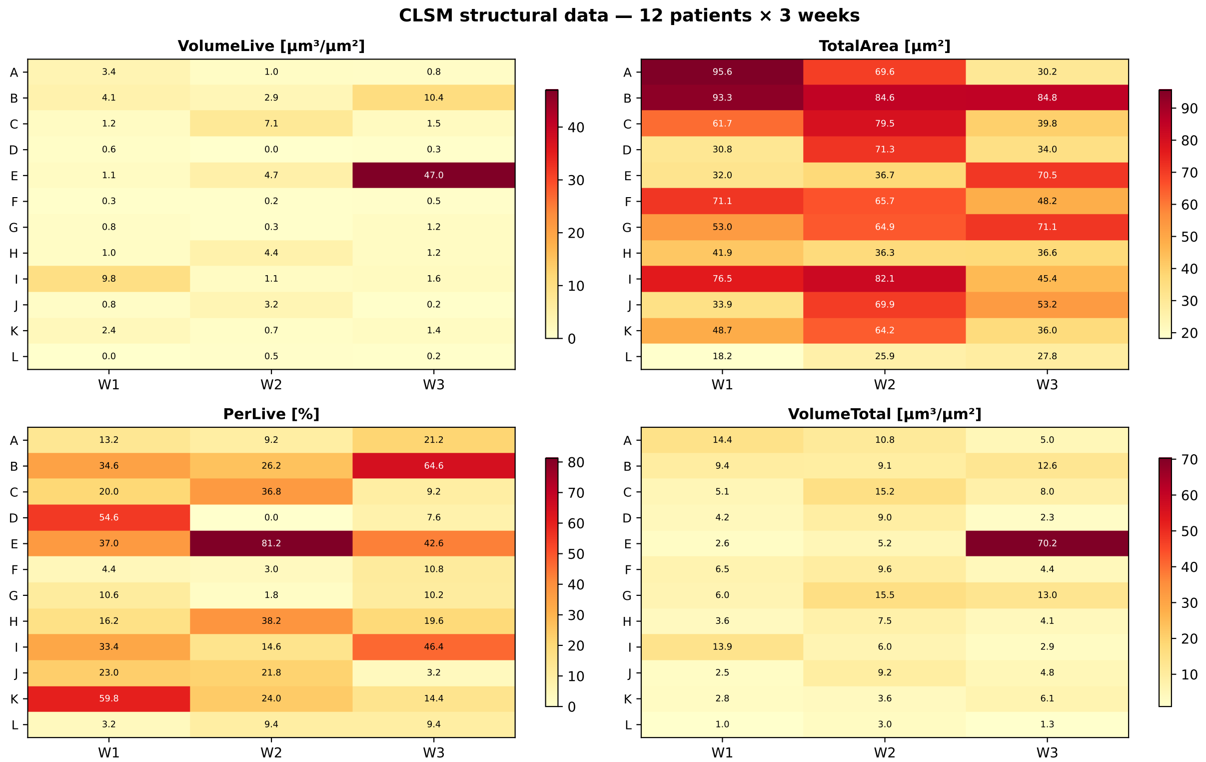

Figure 7: CLSM structural measurements for 12 patients $\times$ 3 weeks. (a) VolumeLive [$\mu m^3\,\mu m^{-2}$]; (b) TotalArea [μm^2]; (c) PerLive [%]; (d) VolumeTotal. Patient I and J were excluded from the ODE analysis (no 16S data) but structural data are shown for completeness. Occupancy $\tilde{\omega}$ used in Eq. (4) is derived from panels (a,b).

4.8 Community type stratification

Hierarchical Ward clustering of the 10 patients by mean 10-guild composition identifies two community types (CTs) with silhouette score 0.59 (Fig. 13).

CT1 (patients D, E, G, K, L): Bacilli-dominated (85±6%), with low Actinobacteria ($<$ 10%). Streptococcal primary colonisers rapidly fill the available niche, leaving little room for competing lineages.

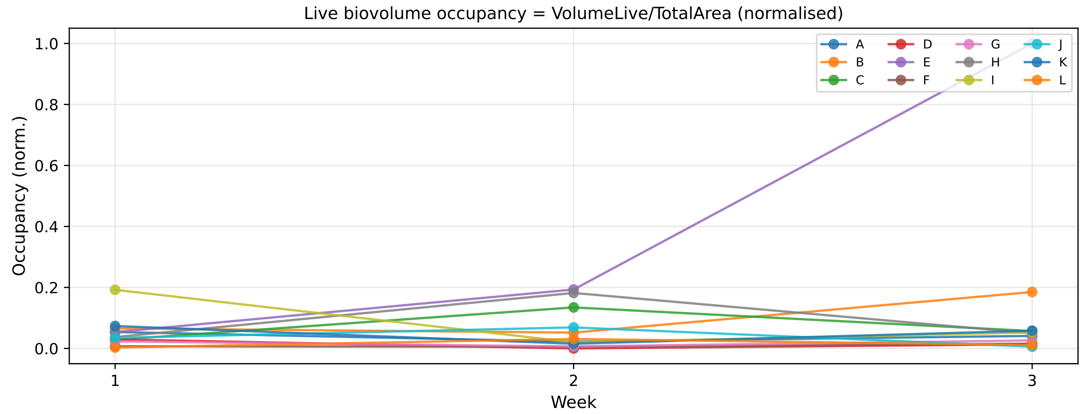

Figure 8: Normalised live-biovolume occupancy $\tilde{\omega}_{p,w} = \omega_{p,w}/\max\omega$ per patient over weeks 1–3. Each line is one patient; occupancy generally peaks at week 1 and declines, consistent with early-colonisation dynamics.

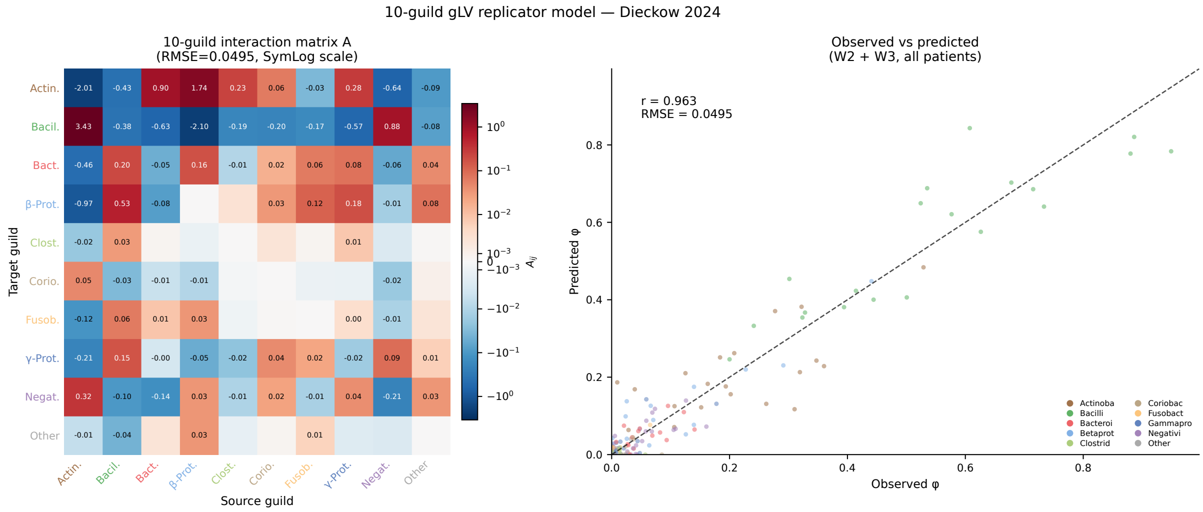

Figure 9: Left: inferred 10×10 guild interaction matrix (red = positive, blue = negative). Right: observed vs predicted relative abundance for weeks 2 and 3 ($r = 0.963$, RMSE = 0.050).

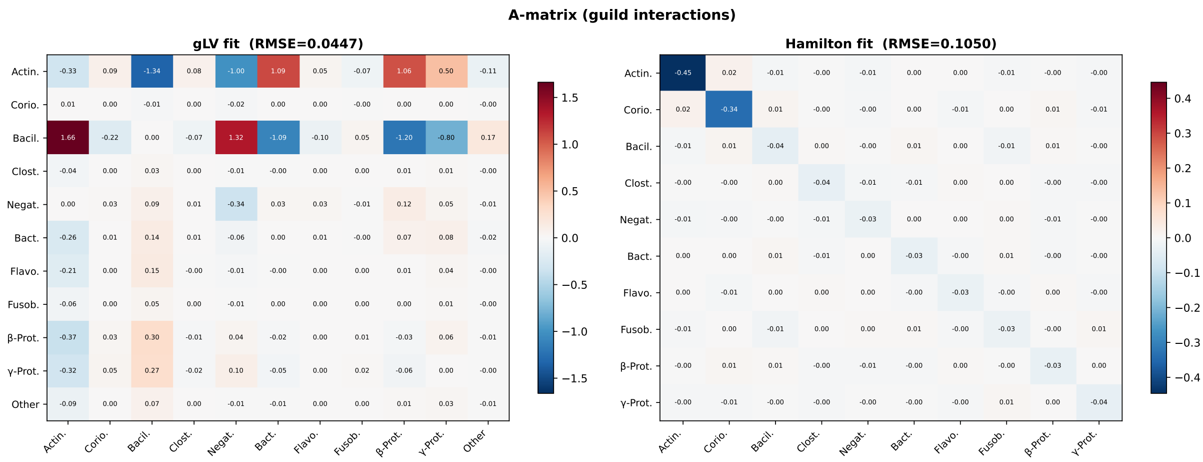

Figure 10: A-matrix comparison: gLV (left, RMSE = 0.050) vs Hamilton ODE with CLSM-informed parameters (right, RMSE $\approx$ 0.061, preliminary). Both models agree on the dominant interactions; the Hamilton ODE yields smaller off-diagonal magnitudes due to the additional mechanistic constraints imposed by the c/α parameterisation.

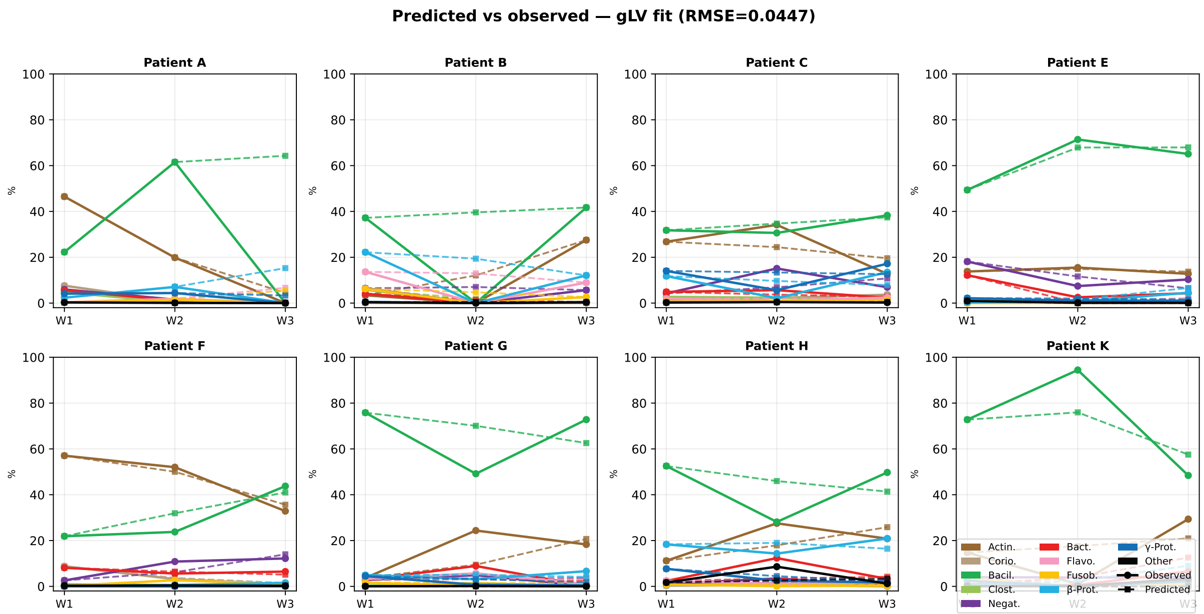

Figure 11: Per-patient predicted (dashed, squares) vs observed (solid, circles) guild trajectories for weeks 1–3. gLV MAP predictions; guild colours as in Fig. 6.

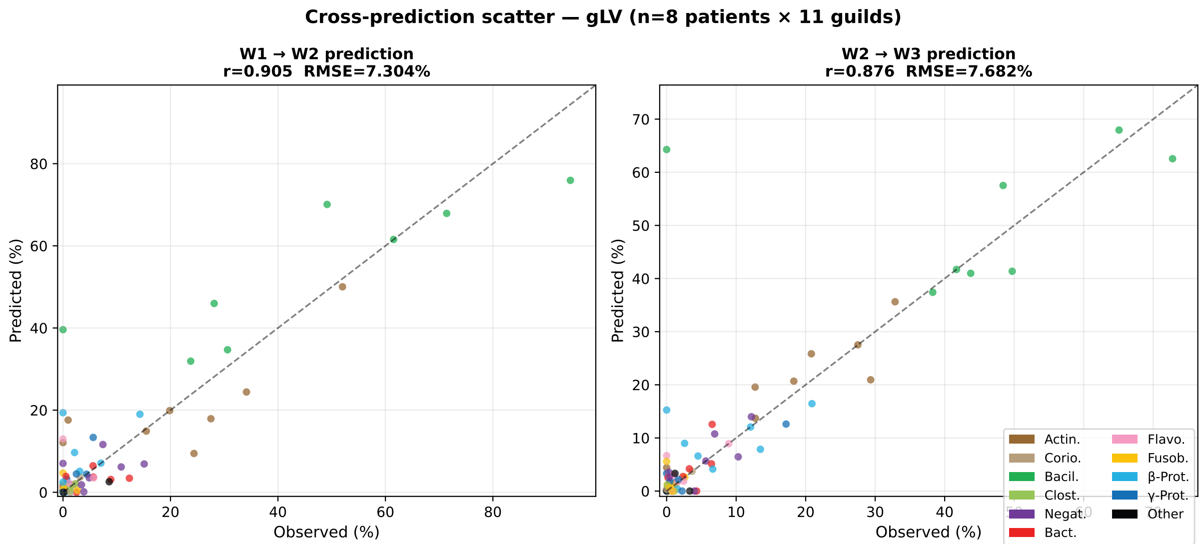

Figure 12: Cross-prediction scatter: observed vs predicted guild fractions (%) for W1→W2 (left) and W2→W3 (right). Each point is one guild × patient; colour indicates guild class. Dashed line: perfect prediction.

CT2 (patients A, B, C, F, H): More diverse; Bacilli reduced (65 ± 12%), Actinobacteria elevated (25 ± 8%), Betaproteobacteria (*Neisseria* family) visible in several samples. Patient F shows the most extreme CT2 phenotype, with Actinobacteria exceeding 60% at week 1.

This dichotomy mirrors the "patient-specificity" reported by Dieckow et al., and is consistent with the community types identified in health-associated vs mucositis peri-implant biofilms in the Szafrański et al. integrated ecosystem analysis [11]. CT1 patients may correspond to a Streptococcus-dominant pioneer community, while CT2 reflects delayed or disrupted streptococcal dominance with stronger Actinomyces and Proteobacteria co-colonisation.

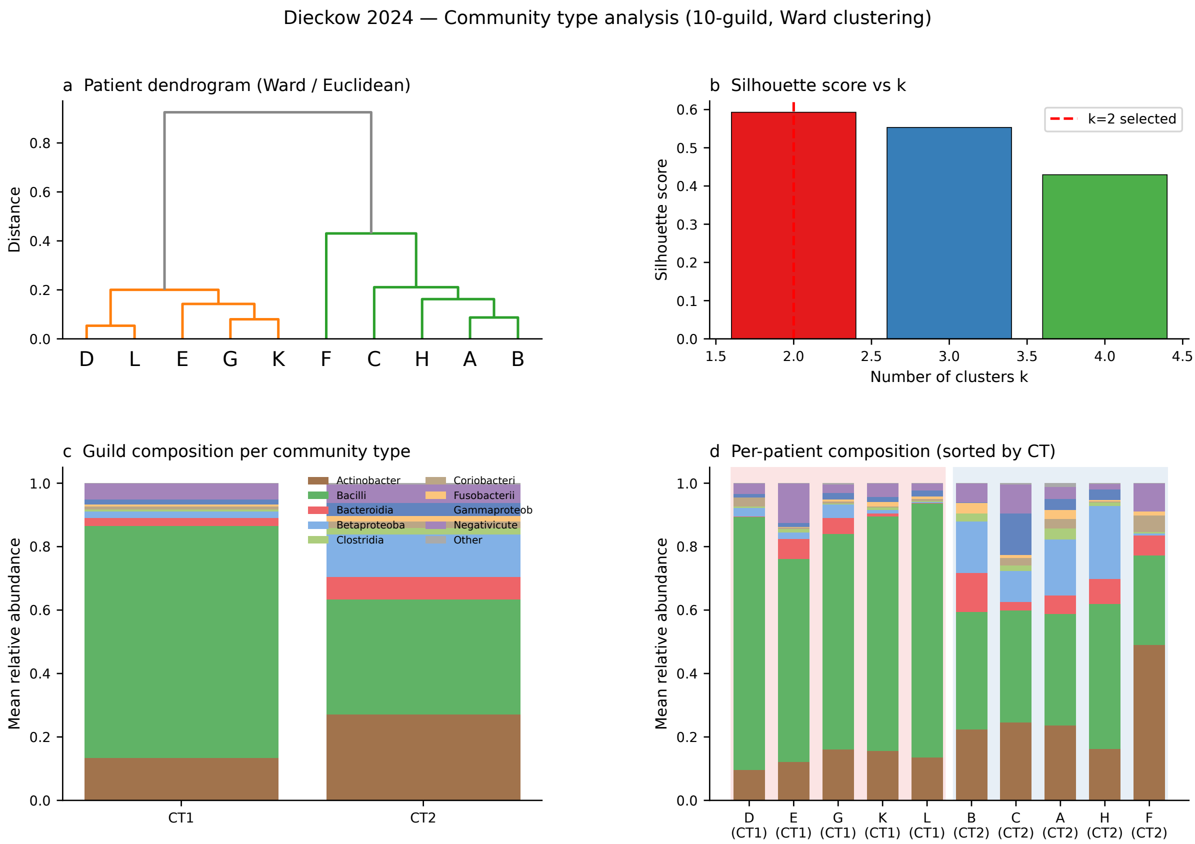

Figure 13: Community type analysis. (a) Ward dendrogram of 10 patients by mean 10-guild composition. (b) Silhouette scores; $k = 2$ selected. (c) Mean guild composition per CT. (d) Per-patient composition sorted by CT; background shading indicates CT assignment.

5 Discussion

Shared interaction topology. The cross-prediction result (Table 2) implies that the ecological interaction network of peri-implant biofilm is substantially conserved between controlled in-vitro and clinical in-vivo settings. The dominant interactions—So–An co-aggregation, Fn–Pg cooperative proteolysis, Vd–Pg pH facilitation—are positive in every condition tested, confirming their status as robust ecological mutualities. This robustness supports the use of in-vitro calibrated models as priors for in-vivo prediction.

An–Vd sign reversal and the V. dispar/V. parvula difference. The most informative discrepancy between in-vitro and in-vivo fits is the negative An–Vd coefficient in the Dieckow data. *V. parvula* is an obligate anaerobe that ferments lactate/succinate produced by *A. naeslundii*, establishing a clear mutualistic relationship in vitro. *V. dispar*, the dominant Veillonella representative in the Dieckow cohort, grows more slowly on lactate and competes with *A. naeslundii* for available carbon sources under in-vivo microaerobic conditions [6]. The Bergey sign prior, which assigns a positive expected sign to An–Vd based on the Dieckow SI metabolic network, is overridden by the data likelihood, suggesting the reversal is not an artefact of limited data.

Patient heterogeneity is primarily growth-rate driven. The patient-specific b estimates span a 15-fold range for individual species, while the shared A converges to a stable MAP. This decomposition is clinically interpretable: differences in salivary flow rate, mucosal immune status, implant surface properties, and initial colonisation seeding all influence how fast each species grows, but the ecological rules governing inter-species interactions are more conserved. Patient F is an outlier with very high estimated growth rates for So, Vd, and Fn, possibly reflecting an unusually rapid biofilm maturation trajectory or sequencing artefacts.

Community types and the guild-level interaction structure. The two community types identified by Ward clustering (CT1: Bacilli-dominant; CT2: Actinobacteria-enriched) reveal a patient-level heterogeneity that is invisible to the five-species Hamilton ODE, which collapses the two CTs into comparable $\hat{\varphi}_{\mathrm{So}}$ values because Bacilli and Actinobacteria are both lumped into the early-coloniser pool. The 10-guild gLV model captures this distinction directly: the inferred Actinobacteria→Bacilli interaction ($A = +0.38$) is the dominant off-diagonal entry, consistent with the textbook role of *Actinomyces* as a bridging organism that co-aggregates with streptococci via type-2 fimbriae [10]. The negative Betaproteobacteria→Actinobacteria coefficient ($A = -0.20$) is biologically plausible: *Neisseria* spp. produce hydrogen peroxide and bacteriocin-like inhibitory substances that suppress *Actinomyces* growth under aerobic early-colonisation conditions [10]. That CT2 patients (higher Betaproteobacteria) show lower Actinobacteria over time is consistent with this mechanism.

The guild model achieves $r = 0.963$ ($\mathrm{RMSE} = 0.050$) in predicting week-2 and week-3 compositions from week-1 initial conditions—comparable to the five-species Hamilton self-fit RMSE of 0.101 on a different (in-vitro) dataset. This demonstrates that a class-level aggregation retains most of the predictive signal while covering 100% of sequenced reads.

Metabolic potential vs. realised dynamics: comparison with Supp. File 1. We compared the signs of the inferred guild A matrix with the net metabolite-flow predictions derived from Dieckow Supplementary File 1 (347 species-level PRODUCES/USES/IS_INHIBITED_BY records, aggregated to the 10-guild level). Of 18 guild pairs with at least one Supp. File 1 prediction, 8 (44%) showed sign agreement and 10 (56%) disagreed.

Agreements include the two most ecologically important interactions: Actinobacteria→Bacilli (+, co-aggregation mediated by fimbrial adhesins and polysaccharide complementarity) and Negativicutes→Bacilli (+, lactic-acid cross-feeding from *Streptococcus* to *Veillonella*).

The most informative disagreements involve Betaproteobacteria (*Neisseria*) and Bacteroidia (*Prevotella*/*Porphyromonas*) acting as net negative regulators of Bacilli in the temporal dynamics ($A = -0.17$ and -0.05 respectively), whereas Supp. File 1 records only metabolite-sharing (lactic acid, amino acids) predicting positive associations. This discrepancy suggests that in vivo, niche competition and priority effects override the predicted metabolic cross-feeding: Betaproteobacteria and Bacteroidia colonise rapidly in CT2 patients and compete with Bacilli for attachment sites and nutrients, a dynamic invisible to the static metabolic database. Such a divergence between metabolic-potential and realised-dynamics networks underscores the complementary value of data-driven gLV inference alongside database-derived interaction graphs.

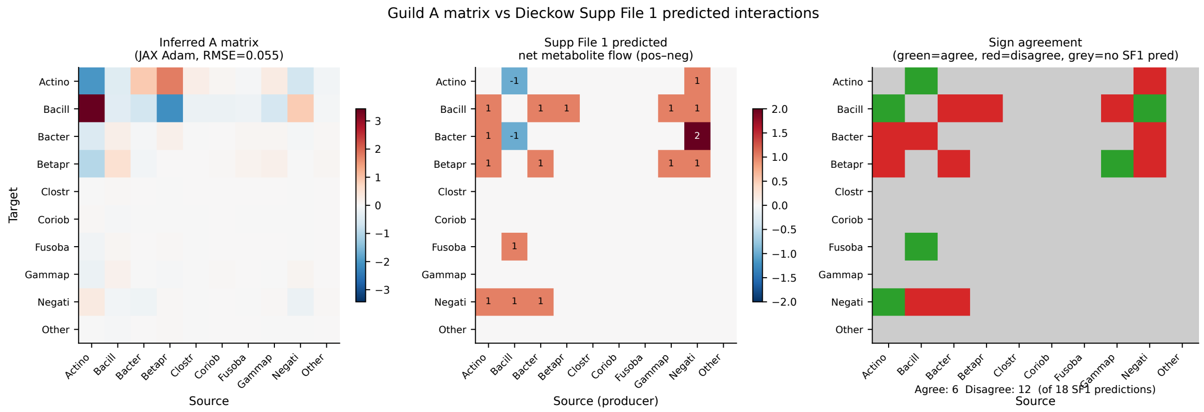

Figure 14: Comparison of inferred guild A matrix with Dieckow Supp. File 1 predicted interactions. Left: inferred A matrix (L-BFGS-B, RMSE = 0.050). Centre: net metabolite-flow score from Supp. File 1 (positive = net PRODUCES→USES; negative = net PRODUCES→IS_INHIBITED_BY). Right: sign agreement (green = agree, red = disagree, grey = no Supp. File 1 prediction). Of 18 predicted pairs, 8 agree and 10 disagree.

Limitations. The Hamilton ODE assumes well-mixed conditions and does not capture the spatial stratification observed in CLSM images. With only three time points, the identifiability of all 65 parameters depends on the shared-A constraint; releasing A per-patient would make the problem underdetermined. The aggregation of 16S data to five model species discards the rich taxonomic resolution of full-length sequencing; the 10-guild extension recovers 100% of reads but collapses within-class diversity. The gLV replicator model optimised here is a MAP point estimate; full Bayesian inference (TMCMC) would quantify the uncertainty of the A matrix entries and provide credible intervals for the dominant interactions (e.g. Actinobacteria→Bacilli, $A = +0.38$).

6 Conclusions

We have extended the Hamilton-TMCMC framework to in-vivo clinical data, jointly inferring a shared interaction matrix and patient-specific growth rates for a 10-patient peri-implant biofilm cohort. The key findings are:

1. **Conserved interaction backbone.** Heine in-vitro A matrices applied to Dieckow data (with patient-specific b) achieve RMSE within 0.002 of the in-vivo self-fit, confirming that ecological interactions are largely conserved across in-vitro and in-vivo settings.
2. **Growth-rate heterogeneity.** Patient variability is captured by patient-specific b; the shared A is stable and biologically interpretable.
3. **Strain-dependent sign reversal.** The negative An–Vd coefficient in the in-vivo fit is consistent with the *V. dispar* (in-vivo) vs. *V. parvula* (in-vitro) substitution, demonstrating that the model is sensitive to ecologically meaningful strain-level differences.
4. **Robust mutualistic backbone.** Three pairs (So–An, Vd–Pg, Fn–Pg) are positive in all five conditions tested, establishing conserved mutualities of the peri-implant ecosystem.
5. **Two community types.** Ward clustering of the 10-guild composition identifies two reproducible patient phenotypes: CT1 (Bacilli-dominant, 5 patients) and CT2 (Actinobacteria-

enriched, 5 patients), with silhouette score 0.59.

6. **Guild-level gLV.** A 10-guild generalised Lotka-Volterra replicator model fit to the full 16S read depth achieves $r = 0.963$ (RMSE $= 0.050$) in week-2/3 prediction, with Actinobacteria$\to$Bacilli as the dominant positive driver.

Future work. Integration of CLSM spatial data (biofilm volume, coverage, viability) with a PDE extension of the Hamilton model [3] would allow the spatial stratification observed by Dieckow et al. to inform mechanistic growth-rate estimates. Full Bayesian inference (TMCMC) of the 10-guild gLV A matrix would provide uncertainty-quantified interaction strengths and enable model selection between CT subtypes. Correlation of CT membership with clinical parameters (implant surface roughness, probing depth, patient medical history) is a priority for a future clinical cohort study.

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